T E C H N I C A L O P E R A T I O N S M A N U A L

HORIZON 87

Solar System Visualization, Observer Guidance & Mission Analysis Suite

Version 1.0 · June 2026 · Desktop Web Application

NAVIGATION INDEX

Contents

§	Section	Contents
01	System Overview	*Architecture, operational layers, and trust-band framework.*
02	Trust Bands & Claim Boundaries	*Definitions of survey, observer, validation, and mission-screening grades.*
03	System Architecture	*Module map, worker architecture, and inter-component communication.*
04	Time Systems & Epochs	*UTC, JDE, ET/TDB, J2000, and Delta T conversion chain.*
05	Ephemeris Engines	*DE441, DE442s, VSOP analytical fallback, and CSPICE state-vector interface.*
06	Coordinate Frames & Vectors	*Cartesian/spherical conversion, geocentric reduction, angular separation.*
07	Lunar Phase & Eclipse Logic	*Illumination model, solar eclipse geometry, lunar eclipse shadow cones.*
08	Observer Presets & Sky Events	*Altitude, twilight, hour angle, and planet parade scanner modes.*
09	ZENITH Validation Lab	*Residual decomposition, Cartesian/angular miss, configuration fields.*
10	Mission Lab & Porkchop Analysis	*Lambert solver, C3, v-infinity, porkchop grid, and filtering.*
11	3D Telemetry Renderer	*Adaptive exposure, WebGL bodies, ring system, and focus controls.*
12	Storage & Precision Assets	Origin *Private File System kernel management, downloader worker, and offline operation.*
13	V1 Limitations	*Platform, scope, and certification boundaries.*
14	Glossary	*Definitions for all technical terms used throughout this document.*
15	External References	*NASA/JPL Horizons, NAIF CSPICE, and JPL ephemeris documentation.*

SECTION 01

System Overview

Horizon 87 is a browser-native astronomy and preliminary mission-analysis suite. It integrates ephemeris-driven solar system visualization, observer-aware sky-event detection, state-vector validation against external reference services, and a Lambert-solver-based transfer-window lab. Core visualization, local precision computation, and Mission Lab operation run client-side once precision assets are installed; live validation still requires an external Horizons reference request.

The application is structured as a hierarchy of trust layers. Each layer carries its own data sources and claim boundaries. A survey-grade orbit sketch operates on different principles than a validation-grade state-vector dossier, and Horizon 87 makes that distinction explicit throughout its interface.

> **IN PLAIN ENGLISH** Horizon 87 is a web app that shows you where planets are, tells you when to look at the sky, checks its own accuracy against NASA/JPL Horizons reference data, and lets you plot interplanetary transfer routes - all running in a desktop browser.

Operational Layers

Layer	What the Operator Sees	Technical Mechanism
Visualizer	Planets, dates, geocentric and heliocentric views, 3D telemetry.	Coordinates generated from the active ephemeris path and projected into 2D/3D display space.
Observer Guide	Local event times, planet parades, eclipse visibility cards.	Sky vectors filtered through twilight, altitude, solar separation, local timezone, and observer presets.
ZENITH Validation	Residual dossier comparing Horizon 87 to NASA/JPL Horizons.	Local CSPICE state vectors compared against live Horizons VECTORS rows under matched frame, time-scale, and correction conventions.
Mission Lab	Porkchop plots and transfer-window energy maps.	Heliocentric states feed a universal-variable Lambert solver; v-infinity, C3, and delta-v proxies are mapped across departure/arrival date grids.
Forge / Sandbox	Future physics workbench - locked in V1.	Legacy N-body integration code present but not exposed until drift and regression tests are complete.

SECTION 02

Trust Bands & Claim Boundaries

A precision application must not make identical promises across every subsystem. Horizon 87 defines four public trust bands - explicit categories that bound what each module may claim and what operators should expect from its output.

Trust Band	Primary Path	Appropriate Use	Valid Claim
Survey-Grade	Analytical fallback engine with automatic precision where coverage is available.	Fast exploration, wide date scanning, visual discovery, education.	Responsive and useful. Not a formal state-vector certificate.
Observer-Grade	Ephemeris geometry plus altitude, twilight, timezone, and observer-location filters.	Planet parades, eclipse cards, local sky guidance.	Geometric observing guidance. Excludes weather, terrain, and atmospheric extinction beyond implemented approximations.
Validation-Grade	ZENITH DE441 / CSPICE plus live NASA/JPL Horizons comparison.	Reproducible residual dossiers and state-vector checks.	High-trust when target, observer, frame, time scale, and correction convention are matched and recorded in the exported dossier.
Mission-Screening	Heliocentric states plus universal-variable Lambert solver.	Transfer-window exploration and approximate energy comparison.	Preliminary mission analysis. Not a trajectory optimization or spacecraft navigation product.

> **NOTE** *Survey-grade results are not diminished by the existence of validation-grade paths. Each band is designed to be honest, not to imply the others are defective.*

SECTION 03

System Architecture

Horizon 87 is decomposed into discrete modules so that rendering, scanning, storage, precision kernels, and validation do not compete within a single execution context.

Module	Role	V1 Behavior
ui.js	Main controller - date/time, scanners, event cards, observer logic, engine selector, 2D render loop.	Sky-event scanning routes through AstroEngine functions. ZENITH and Precision paths engage where the feature supports them.
precision_engine.js	Local DE442s precision path.	Loads de442s.bsp into a CSPICE/WASM runtime; exposes state-vector and Geometry Finder eclipse windows when ready.
zenith_engine.js	ZENITH DE441 runtime and state-vector bridge.	Worker-first execution with main-thread fallback; exposes async stateVector/calculate and synchronous render paths.
zenith_storage.js	Origin Private File System package manager.	Stores split DE441 kernel files, leap-second kernel, CSPICE JS/WASM, and metadata in the Origin Private File System.
zenith_downloader_worker.js	Large-asset background downloader.	Chunked range downloads and SHA-256 hash verification, isolated from the UI thread.
zenith_worker.js	Background ZENITH state calculator.	Mounts local kernels inside a worker runtime; returns calculated vectors via postMessage.
three_mode.js	3D telemetry renderer.	Textures, tone mapping, focus transitions, flight controls, dynamic exposure, star fields, ring alpha, and HUD.
reloaded_labs.js	Validation and Mission Lab.	Fetches Horizons references, decomposes residuals, exports dossiers, solves Lambert grids, and draws porkchop plots.

Web Worker Communication

A Web Worker is a separate JavaScript execution context. It performs heavy computation without blocking animation, input handling, or rendering on the main thread. Workers communicate exclusively through serialized message passing:

```
main thread  →  postMessage(payload)  →  worker
main thread  ←  postMessage(result)   ←  worker
```

The downloader worker fetches large kernel packages in byte-range chunks and verifies file integrity while the interface remains responsive. The ZENITH worker performs CSPICE state-vector calls without impacting the render loop. Each request carries a unique ID so returned results can be matched to the awaiting Promise.

SECTION 04

Time Systems & Coordinate Epochs

Astronomical computation requires more than a display date. Civil UTC timestamps mark calendar events; ephemeris calculations demand continuous dynamical time arguments that do not accumulate leap-second discontinuities. Horizon 87 maintains separate representations for each context and converts between them at well-defined boundaries.

Julian Ephemeris Day (JDE)

The internal time scalar is formed by offsetting a Unix timestamp to the Julian epoch and applying a Delta T correction:

$$JDE = \frac{unix_{\mathrm{ms}}}{86\,400\,000} + 2\,440\,587.5 + \frac{\Delta T}{86\,400}$$

JDE from Unix milliseconds

Delta T approximates the difference between Earth-rotation-based Universal Time and uniform dynamical time. Its value grows slowly over decades and is sourced from published tables for the current era.

Ephemeris Time (ET / TDB)

SPICE kernels require ephemeris time as continuous seconds elapsed since the J2000 epoch - 1 January 2000, 12:00 TDB:

$$ET = (JDE - 2\,451\,545.0) \times 86\,400$$

Seconds past J2000

VSOP Series Argument (τ)

Analytical VSOP-style ephemeris paths use a Julian-millennia parameter:

$$\tau = \frac{JDE - 2\,451\,545.0}{365\,250}$$

Julian millennia from J2000

Term	Definition
UTC	Civil time scale used for user-facing dates and event labels.
ΔT (Delta T)	Correction from Earth-rotation time to uniform dynamical time. Applied during JDE formation.
JDE	Julian Ephemeris Day - the continuous astronomical time scalar used throughout the application.
TDB / ET	Dynamical ephemeris time used by SPICE-style state-vector calculations. For V1 UI explanations it is close to TT, but validation records the actual query epoch and convention rather than treating time scales as interchangeable.
J2000	Standard reference epoch: 1 January 2000, 12:00 TT/TDB. Zero-point for ephemeris time calculations.

SECTION 05

Ephemeris Engines

An ephemeris is a model or tabulated dataset giving the position and velocity of a celestial body over time. Horizon 87 maintains multiple ephemeris paths because speed, coverage, file size, and validation strength are competing constraints that cannot be simultaneously optimised by a single model.

Engine	Data Source	Best Application	Trust Level
Stargazer / Auto Precision	DE442s SPK kernel via local CSPICE/WASM. Analytical fallback outside kernel coverage.	Default exploration, modern-date visual precision.	Strong visual/observer path where coverage and runtime are ready; fallback is survey-grade.
ZENITH	DE441 split BSP kernel + CSPICE/WASM + leap-second kernel stored in Origin Private File System.	Validation-grade state vectors and residual dossiers.	Kernel-authoritative local DE441 path after installation - provided target, frame, and correction conventions match the reference.
VSOP Analytical Fallback	Trigonometric and Poisson-series orbital theories.	Immediate loading, broad scanning, deep-time continuity, graceful degradation.	Survey and education. Not a formal state-vector certificate.
Forge (Locked)	N-body workbench integration code.	Future physics sandbox.	Locked in V1.

State Vector Interface

When the precision runtime is active, all state-vector calls are routed through a CSPICE-compatible WebAssembly layer. The conceptual call returns a six-component vector - three position and three velocity - relative to a specified observer:

$$state = spkez(target,\ ET,\ frame,\ aberration,\ observer)$$

SPICE-style state-vector call

The application does not implement its own Chebyshev polynomial parser over raw JPL binary data. The SPK file is loaded into the CSPICE-compatible runtime, and that runtime is queried for state vectors. The authority for the vector lies with the kernel and the SPICE toolkit.

> **IN PLAIN ENGLISH** ZENITH lets Horizon 87 use the same family of JPL ephemeris data used by professional astronomy software - stored locally in the browser, with no ongoing server connection required.

SECTION 06

Coordinate Frames & Vector Mathematics

Horizon 87 operates in Cartesian three-space and converts between coordinate representations only at interface boundaries - projection into display space, observer altitude calculations, and angular-separation filters.

Cartesian to Spherical Conversion

$$r = \sqrt{x^2 + y^2 + z^2}$$

Radial distance

$$L = atan2(y,\ x)$$

Ecliptic longitude

$$B = arcsin\left(\frac{z}{r}\right)$$

Ecliptic latitude

Geocentric Vector Reduction

To obtain the Earth-centred position of a target body, the Earth's heliocentric state is subtracted from the target's heliocentric state - both in the same coordinate frame:

$$r_{geo} = r_{planet}(helio) - r_{Earth}(helio)$$

Geocentric reduction

Angular Separation

$$cos\,\theta = sin\,B_1\,sin\,B_2 + cos\,B_1\,cos\,B_2\,cos(L_1 - L_2)$$

Great-circle angular separation

Circular Arc Span

Planet parades require measuring how much of the sky a group of bodies occupies. Naive longitude subtraction fails when the group straddles 360°/0°. The correct formulation identifies the largest gap in the sorted longitude list and subtracts from 360°:

$$span = 360° - max\ circular\ gap(sorted\ longitudes)$$

Circular arc span - handles wraparound correctly

> **EXAMPLE** *Longitudes 355°, 2°, and 8° occupy a 13° arc, not a 353° arc. Circular arc span handles this correctly.*

Frame / Quantity	Used For
Heliocentric vector (J2000)	Solar-system map positions and mission departure/arrival states.
Geocentric vector	Earth-centred sky positions, parade geometry, and observer-facing event logic.
J2000 inertial frame	Validation state vectors and reference comparison against NASA/JPL Horizons.
Ecliptic longitude / latitude	2D display, sky grouping, elongation filtering, and eclipse angular tests.
Equatorial RA / Dec	Observer altitude calculations and local sky conversion.

SECTION 07

Lunar Phase & Eclipse Detection

The Moon module blends user-facing phase display with geometric eclipse detection. A new Moon is not automatically a solar eclipse, and a full Moon is not automatically a lunar eclipse. Horizon 87 requires geometric conditions to be satisfied before classifying an event.

Illumination

$$illumination = 0.5 \times (1 - cos(2\pi \times phase)) \times 100\%$$

Illumination from phase angle

Monthly extrema are hard-pinned to the sampled peak new/full dates to ensure 0% and 100% are displayed only at the correct epochs.

Solar Eclipse Detection

A global solar eclipse is possible only when the Moon's angular separation from the Sun falls inside the combined angular radius and parallax overlap limit:

$$global\ limit = r_{Moon} + r_{Sun} + parallax_{\mathrm{m}}$$

Solar eclipse global geometry limit

$$isSolarEclipse \Leftrightarrow separation < global\ limit$$

Eclipse condition

Lunar Eclipse Detection

For lunar eclipses, the angular miss distance is converted to a physical offset at lunar distance, then compared against the umbral and penumbral cone radii:

$$axis\,offset_{km} = d_{Moon} \times sin(separation)$$

Physical shadow offset

$$r_{umbra} = R_{Earth} - d_{Moon} \times \frac{R_{Sun} - R_{Earth}}{d_{Sun}}$$

Umbral cone radius at Moon

$$r_{penumbra} = R_{Earth} + d_{Moon} \times \frac{R_{Sun} + R_{Earth}}{d_{Sun}}$$

Penumbral cone radius at Moon

Eclipse Tier	Meaning
Global event	The eclipse s - occurring somewhere on Earth or within Earth’s shadow cone.
Local event	The selected observer passes altitude and contact-geometry gates, making the event visible from that site.
Practical horizon	V1 enforces a conservative altitude floor, suppressing very low-elevation events from being labelled accessible.
GF-first path	When CSPICE Geometry Finder is available, it identifies candidate eclipse windows before UI geometry filtering.
Fallback sweep	If Geometry Finder is unavailable, coarse-to-fine angular separation scans with peak refinement are used instead.

SECTION 08

Observer Presets, Local Time & Sky Events

Observer features translate mathematical sky coordinates into human-facing guidance. A planet positioned above the ecliptic plane is not necessarily visible from a specific location at a useful hour. Horizon 87 applies altitude, twilight, timezone, and solar-separation filters before presenting any sky-event result.

Altitude Calculation

$$\sin(alt) = \sin\varphi \sin\delta + \cos\varphi \cos\delta \cos H$$

Altitude from observer coordinates

Hour Angle

$$H = GMST + longitude - RA$$

Local hour angle

Civil Twilight

$$\cos H_{twi} = \frac{\sin(-6°) - \sin\varphi \sin\delta_{\odot}}{\cos\varphi \cos\delta_{\odot}}$$

Civil twilight hour angle

Planet Parade Scanner

The parade scanner distinguishes three increasingly constrained event classes:

Mode	Tests Applied	Not Included
Tight Alignment	A subset of planets fits within a requested circular arc after solar-glare filtering.	Altitude, twilight, weather, or local horizon guarantee.
Geometric Parade	Enough planets are on one side of the Sun, grouped within the maximum arc and outside the glare exclusion zone.	Full local observing guarantee.
Observable Parade	At civil twilight, sufficient planets pass altitude, elongation, and span gates for the selected observer.	Weather, terrain, obstructions, optics, sky brightness beyond heuristics.

$$elongation = angular\ separation(planet,\ Sun)$$

Solar elongation

$$visible \iff altitude \geq 5° \ AND \ elongation \geq min\ separation$$

Observable acceptance condition

SECTION 09

ZENITH Residual Validation Lab

The ZENITH validation module is the highest-trust scientific component in Horizon 87 V1. It compares local CSPICE state vectors against live NASA/JPL Horizons VECTORS references. The comparison carries scientific weight because the target, observer, coordinate frame, time scale, and aberration correction are explicitly matched and recorded in the exported dossier.

> **IN PLAIN ENGLISH** Horizon 87 asks NASA/JPL Horizons for a reference state vector, compares it against the state computed locally, and reports the difference. In selected matched validation comparisons, that difference has been far smaller than the width of a human hair.

Residual Decomposition

$$\Delta r = r_{ZENITH} - r_{Horizons}$$

Cartesian residual vector

$$total\,residual_{\mathrm{m}} = |\Delta r| \times 1000$$

Total residual in meters

$$\Delta r_{radial} = (\Delta r \cdot \hat{r}_{ref})$$

Radial / range residual

$$\Delta r_{tangential} = \sqrt{|\Delta r|^2 - {\Delta r_{radial}}^2}$$

Perpendicular / tangential residual

Angular Residual

$$angular\ residual'' = \frac{\Delta r_{tangential}}{|r_{ref}|} \times 206\,264.806\,247$$

Angular residual in arcseconds

Configuration Field	Definition
Frame	J2000 inertial frame for raw state-vector certificate comparisons.
Observer	Earth geocenter (Horizons identifier 500@399) for validation fixtures.
Correction NONE	Raw geometric vector comparison - the cleanest and most reproducible certification convention.
LT / LT+S	Light-time and stellar-aberration diagnostic modes; meaningful only when matched to the reference convention.
ET sweep	Microsecond-offset diagnostic checking whether a small timing adjustment improves the residual.
Strict Mode	V1 record and display posture. Pass/fail thresholds are read from the exported dossier: sub-metre, 10 m, and 50 m bands.

VALIDATION CLAIM *In matched ZENITH-vs-NASA/JPL Horizons validation runs, Horizon 87 has demonstrated micrometer-scale residual agreement for selected state-vector comparisons. This applies to run-specific results under matched conventions. Quote exact values from the exported dossier - including timestamp, kernel version, frame, observer, correction convention, fixture count, and maximum residual.*

SECTION 10

Mission Lab & Porkchop Analysis

The Mission Lab is a first-order interplanetary transfer screening tool. It samples departure and arrival date combinations, retrieves heliocentric state vectors, solves Lambert's problem for each cell, and maps the resulting energy metrics as a colour-coded porkchop plot.

Transfer Geometry

$$TOF = t_{arrival} - t_{departure}$$

Time of flight

$$v\infty_{dep} = |v_{transfer,dep} - v_{body,dep}|$$

Departure hyperbolic excess speed

$$v\infty_{arr} = |v_{transfer,arr} - v_{body,arr}|$$

Arrival hyperbolic excess speed

$$C3 = {v\infty_{dep}}^2$$

Characteristic launch energy (km^2/s^2)

$$total\ v\infty = v\infty_{dep} + v\infty_{arr}$$

Combined energy metric for window ranking

Universal-Variable Lambert Solver

The Lambert problem asks: given two position vectors and a transfer time, find the orbit that connects them. Horizon 87 uses the universal-variable formulation with Stumpff functions:

> **IN PLAIN ENGLISH** Pick a departure date and an arrival date. The Mission Lab finds the trajectory that connects the selected origin and target in that time, then colours the result by how much energy the trip would cost. Lower-energy cells indicate cheaper transfer windows.

$$A = sin(\Delta\theta) \times \sqrt{\frac{r_1 r_2}{1 - cos\,\Delta\theta}}$$

Auxiliary variable A

$$t = \frac{\left(\frac{y}{C}\right)^{3/2} \times S + A\sqrt{y}}{\sqrt{\mu}}$$

Time-of-flight from universal variable

Feature	Description
Date grid	Each cell represents one departure date paired with one arrival date.
Lambert solve	Finds the conic arc connecting two heliocentric position vectors in the requested transfer time.
Finite-difference velocities	Body velocities estimated from adjacent state samples; subtracted from transfer velocities to obtain v-infinity.
Percentile normalisation	Colours normalised by percentile so a single outlier does not compress the visible energy basin.
Contour filtering	Contours exclude invalid cells and branch discontinuities to suppress artefacts at mathematical branch cuts.
Stable-best filter	Rejects suspicious isolated minima and boundary artefacts before presenting the best candidate window.

> **SCOPE** *Mission Lab outputs are preliminary Lambert metrics. They do not account for launch vehicle performance, gravity-assist stages, finite-burn arcs, or operational navigation constraints.*

SECTION 11

3D Telemetry Renderer

The 3D mode is a visualisation layer, not an independent ephemeris authority. It receives position data from the active precision path and projects it into a textured, navigable WebGL scene. Its scientific value lies in interpretability: scale relationships, relative orbital motion, phase angles, and spatial context that are difficult to convey in a 2D projection.

Adaptive Exposure

$$E_{target} = clamp\left(0.82 \times \sqrt{\frac{0.24}{L_{metered}}},\ 0.055,\ 1.85\right)$$

Adaptive exposure from metered luminance

This keeps the Sun, dark planetary surfaces, and night-hemisphere features simultaneously readable within a single frame.

Rendered Features

- Textured spheres for major bodies; Earth includes cloud, night-side, specular, and normal maps.
- Saturn ring system with alpha transparency texture and slow rotational drift.
- Moon orbit rendered with inclination and evolving orbital terms for visual plausibility.
- Double-click focus transitions using eased interpolation; camera follows the focused body during motion.
- Frame-gap clamping for long idle periods or tab-background pauses - prevents discontinuities from rendering lag.
- Star field background calibrated for interplanetary distance scales.

> **NOTE** *Every drawn frame uses the active ephemeris path, which may be warming up, cached, or falling back to the analytical engine. The 3D renderer is not a separate validation certificate.*

SECTION 12

Storage, Installation & Precision Assets

ZENITH and the precision runtime depend on large binary kernel files managed through the Origin Private File System - Horizon 87's internal name for its precision-asset storage layer. It uses the browser's Origin Private File System (OPFS) to store DE441/DE442s kernels, CSPICE runtime files, leap-second data, and package metadata locally on the operator's device, scoped to the website origin and inaccessible to other sites or applications. Once installed, precision assets persist across sessions without re-download.

Origin **Private File System**

The Origin Private File System is Horizon 87's internal name for its precision-asset storage layer. It uses the browser's Origin Private File System to store DE441/DE442s kernels, CSPICE runtime files, leap-second data, and metadata locally on the user's device. No external server access is required once installation is complete.

Asset	Purpose
DE441 split BSP files	Large ZENITH validation-grade planetary and lunar ephemeris kernel, split for progressive download.
DE442s BSP	Compact precision-runtime kernel for Stargazer/PrecisionEngine coverage window.
naif0012.tls	Leap-second kernel required by SPICE time-handling functions.
cspice.js / cspice.wasm	Compiled CSPICE bridge running inside the browser via WebAssembly.
Metadata JSON	Package inventory including file sizes, SHA-256 hashes, and installation status.

IN PLAIN ENGLISH On first use, Horizon 87 downloads JPL ephemeris files to your browser's private storage - similar to an app cache but persistent. After that, all precision calculations run locally with no internet connection required, except for live validation.

The downloader worker fetches kernel files in byte-range chunks, writing directly to the Origin Private File System. SHA-256 integrity checks are performed on completion where metadata provides expected hashes.

> **OFFLINE OPERATION** *Once all precision assets are installed, ZENITH and PrecisionEngine operate fully offline. The only remaining network dependency is the live Horizons reference fetch within the ZENITH Validation Lab.*

SECTION 13

V1 Limitations & Known Constraints

Platform

- Requires a modern desktop browser with WebAssembly support and Origin Private File System API access for precision asset installation. Mobile browsers are not supported in V1.

Affiliation & Certification

- Horizon 87 is not affiliated with, endorsed by, or certified by NASA, JPL, or any space agency. NASA/JPL Horizons and NAIF SPICE are external reference ecosystems used for comparison and kernel access.
- Micrometer-scale residual results are specific to validation runs under matched conventions - not a universal promise for every object, time, or display mode.

Sky & Observer Features

- Sky scanners are survey-grade and observer-grade tools. They do not produce formal state-vector certificates.
- Local visibility predictions exclude weather, terrain, buildings, atmospheric extinction beyond implemented approximations, and individual eyesight variation.

Mission Analysis

- Mission Lab outputs are preliminary Lambert transfer metrics and do not include launch vehicle performance, gravity-assist stages, finite-burn arcs, or operational navigation inputs.

Forge / Sandbox

- The Forge/Sandbox N-body workbench is present in the codebase but locked from public access in V1 pending energy-drift bounds and regression coverage.

Rendering vs. Validation

- ZENITH visual rendering, ZENITH state-vector validation, and PrecisionEngine eclipse windows are related but distinct code paths with independent readiness states and fallback Behavior

SECTION 14

Glossary

Term	Definition
Aberration Correction	SPICE/Horizons option governing whether raw geometric vectors or apparent vectors incorporating light-time and stellar-aberration effects are returned.
Angular Miss	The perpendicular component of a vector residual, expressed as a physical offset distance or angular error on the sky.
BSP / SPK	Binary SPICE Planetary Kernel - the file format used to store object positions and velocities as a function of time.
C3	Characteristic launch energy. Equals the square of departure v-infinity (km^2/s^2). Standard metric for comparing launch difficulty.
CSPICE	C-language implementation of the NAIF SPICE Toolkit, used to access kernel data and compute state vectors.
DE441 / DE442s	JPL Development Ephemeris kernel families providing high-accuracy planetary and lunar positions and velocities.
ET / TDB	Ephemeris Time / Barycentric Dynamical Time - the continuous dynamical time argument used in SPICE-style calculations.
J2000	Standard inertial reference frame and epoch: 1 January 2000, 12:00 TT/TDB.
JDE	Julian Ephemeris Day - the continuous astronomical time scalar used throughout the Horizon 87 pipeline.
Lambert Problem	The orbital mechanics problem of finding a conic transfer arc connecting two position vectors in a specified time.
Origin Private File System	Horizon 87's internal name for its precision-asset storage layer, built on the browser's Origin Private File System (OPFS).
Porkchop Plot	A departure-vs-arrival date grid coloured by mission energy or time-of-flight metric, used to identify optimal launch windows.
Residual	The difference between a locally computed state vector and the corresponding reference value from an external source.
State Vector	A six-component vector - three position coordinates and three velocity components - fully specifying an object's instantaneous kinematic state.
V-infinity ($v\infty$)	Hyperbolic excess velocity - the speed of a spacecraft relative to a body at the edge of that body's gravitational sphere of influence.
Web Worker	A background JavaScript execution thread communicating with the main UI thread via message passing, used to offload heavy computation.

SECTION 15

External References

Resource	URL
NASA/JPL Horizons API Documentation	https://ssd-api.jpl.nasa.gov/doc/horizons.html
NASA/JPL Horizons System	https://ssd.jpl.nasa.gov/horizons/
NAIF CSPICE SPK Required Reading	https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/req/spk.html
NAIF SPICE Kernel Required Reading	https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/req/kernel.html
JPL Planetary Ephemeris Export Page	https://ssd.jpl.nasa.gov/planets/eph_export.html
JPL DE440 / DE441 Overview	https://ssd.jpl.nasa.gov/doc/de440_de441.html